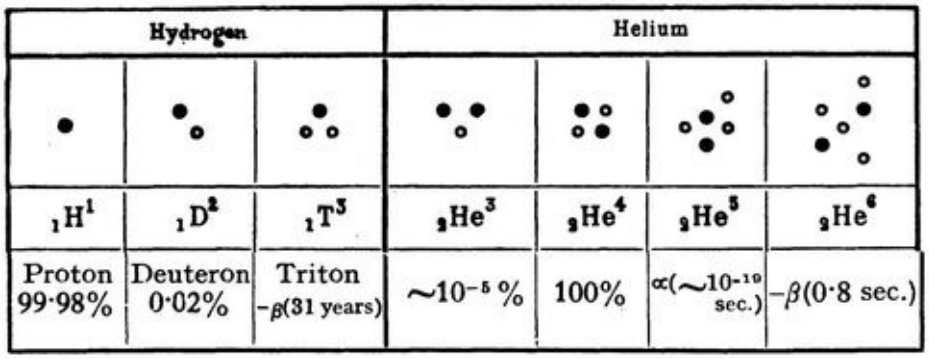
isotope of hydrogen, which consists of one proton and one neutron. This isotope occurs in natural hydrogen to the very small extent of 0·02%. This variety of hydrogen is called *deuterium,* and its nucleus a *deuteron.* Since it differs chemically in certain respects from ordinary hydrogen, it is usually designated by the symbol D, or more exactly, $_1D^2$. It may be written also as $_1H^2$. This is the so-called 'heavy hydrogen'. A third variety of hydrogen was discovered later; its nucleus consists of 1 proton and 2 neutrons. It is called *tritium,* and its nucleus is referred to as a *triton.* Its symbol is $_1T^3$ or $_1H^3$. This nucleus is not stable; it is radioactive, with a very long half-life (probably about thirty-one years), and is an emitter of electrons. Therefore, tritium does not occur in nature, but is a product of nuclear transmutation processes.



The next simplest element is helium, the nucleus of which contains two protons. Helium occurs also in the form of several isotopes, each with a different number of neutrons in the nucleus. The lightest helium nucleus consists of 2 protons and 1 neutron, and its symbol, therefore, is $_2He^3$. It became known as the product of the radioactive transmutation of the triton, $_1T^3$, although it occurs also in natural helium, in very minute, quantities. (This nucleus and the nuclei of the other helium isotopes, are shown in the above chart.) The next nucleus is that of ordinary helium, consisting of 2 protons and 2 neutrons; its symbol, as we already know, is $_2He^4$. This is an especially stable structure. There are two other helium nuclei, neither of which is stable. They contain, respectively, 3 and 4 neutrons, and are therefore written $_2He^5$ and $_2He^6$. They are not present in natural gaseous helium.

Continuing to add further protons and neutrons, we come to nuclei which are increasingly complex in structure. We could make a chart of all the existing nuclei, recording their atomic numbers—in other words, the number of their protons—designated by the symbol *Z* as abscissa, and the numbers of their neutrons, designated by the symbol *N*, as ordinate. It is however more convenient for the printer to adopt a different system in which the abscissae,